ANGLO-CHINESE JUNIOR COLLEGE
Preliminary Examination 2016

BIOLOGY **9648/01**

HIGHER 2 **31 August 2016**

Paper 1 Multiple Choice **1 hour 15 mins**

Additional Material: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, pencil clips, highlighters, glue or correction fluid.
Write your name, centre number and index number on the Answer Sheet provided.

There are **forty** questions in this paper. Answer **all** questions. For each question there are four possible answers, **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate answer sheet.

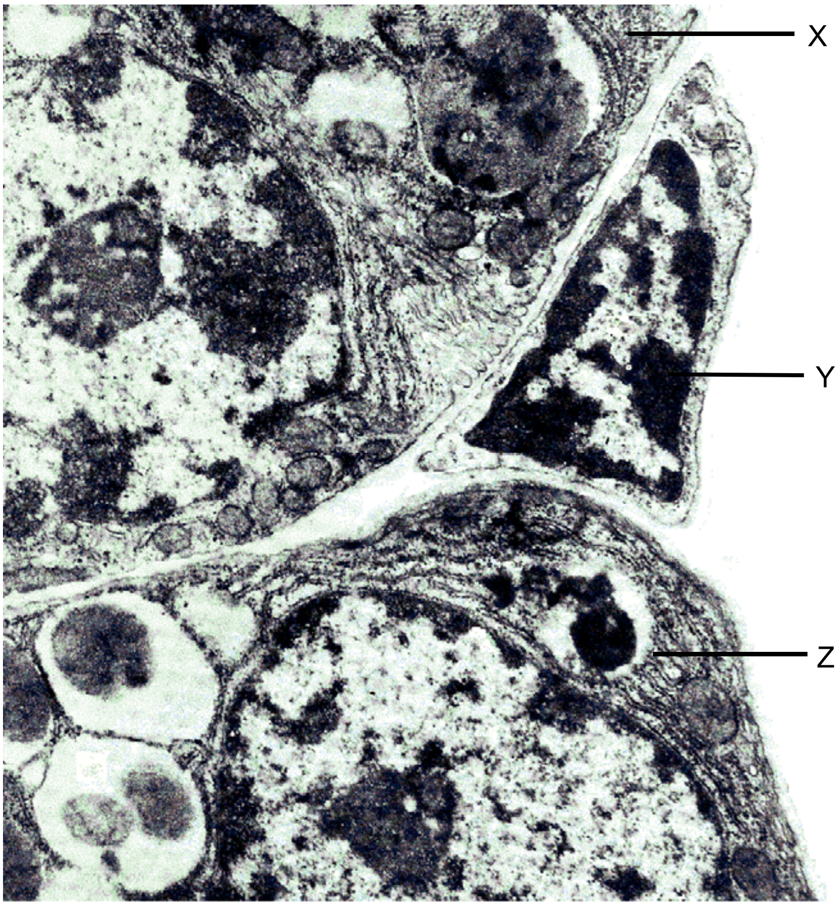
Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

Calculators may be used.

This question paper consists of **27** printed pages.

1 The following electron micrograph shows three adjacent cells, X, Y and Z.



Which of the following descriptions about these cells is **not** true?

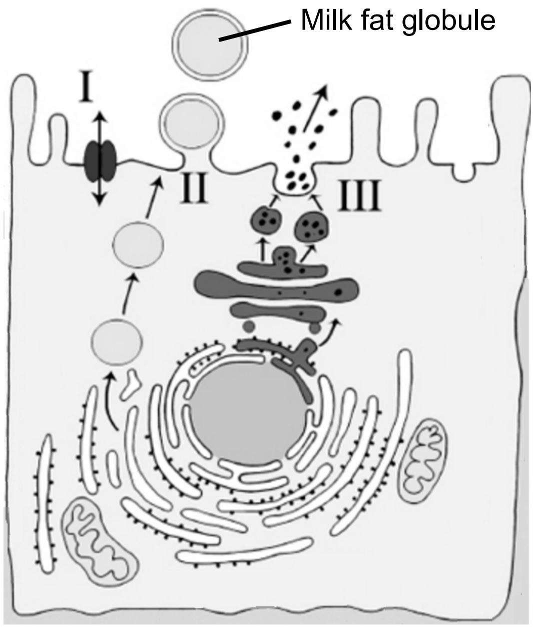
A Cell X contains both linear and circular molecules as its genetic material.

B Cell Y has a rigid cellulose cell wall which resists osmotic lysis.

C Cell Z contains 40S and 60S ribosomal subunits in its cytoplasm.

D Both cell X and cell Z possess intracellular membranes.

2 The diagram shows an epithelial cell in the mammary glands of a mammal. Such cells are responsible for the secretion of milk, an emulsion made up of lactose, lipids, proteins, ions and water. The various substances in milk are secreted through three different transport processes I, II and III.



Which of the following correctly describes the secretion of substances in milk?

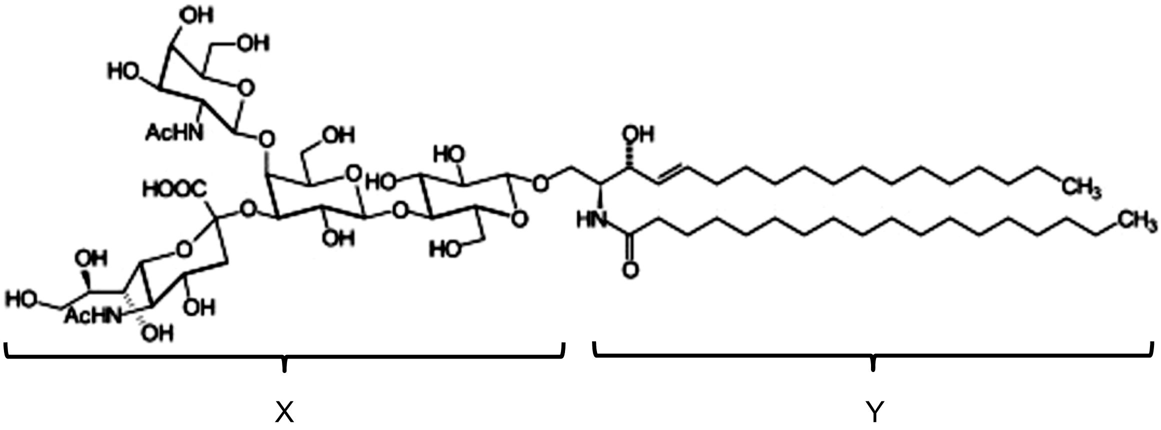
A The secretion of large fat globules occurs by exocytosis, with the expenditure of ATP.

B Lactose and ions have to be secreted through process I due to their hydrophobicity.

C Water can be transported in vesicles budding from the rough endoplasmic reticulum and secreted through process II.

D Milk proteins are transported out of the cell through process III, due to their large molecular size.

3 A ganglioside is a molecule commonly found in cell membranes, and its structure comprises two main components, X and Y.



Which of the following statements regarding a ganglioside is true?

A It comprises two fatty acid chains joined to a glycerol molecule by ester bonds.

B Component X helps to regulate the permeability of the cell membrane.

C Component X is responsible for cell-to-cell recognition and acts as a receptor for other molecules.

D Component X is embedded in the cell membrane while component Y faces the extracellular fluid.

4 The winged bean is a tropical crop that has high protein content. Winged beans have been reported to have a low level of protein digestibility. Protease inhibitors in the bean have been suggested to be responsible for the low digestibility.

In an experiment to study the effect of heat treatment on protein digestibility in winged beans, one of two winged beans was subjected to heat treatment. Trypsin was subsequently added to each reaction mixture and incubated for 30 minutes. The protein concentration of each reaction mixture at the beginning and at the end of the incubation period is shown in the table below.

Incubation period / min	Protein concentration of the reaction mixture / %	
	Trypsin + heat-treated winged bean	Trypsin + untreated winged bean
0	100	100
30	40	70

Which of the following statements is a likely explanation for the data shown?

A Heat treatment of winged bean caused the activation of trypsin inhibitors.

B Heat treatment of winged bean denatured trypsin by changing the 3-dimensional configuration of the enzyme.

C Heat treatment of winged bean disrupted cellular structure and improved accessibility of trypsin to protein.

D Heat treatment of winged bean lowered the activation energy of trypsin and increased the rate of enzyme-catalysed reaction.

5 In *Caenorhabditis elegans*, studies on the synapsis of homologous chromosomes revealed that one end of each chromosome becomes attached to protein patches on the nuclear envelope. The protein patches form a bridge between the chromosomes and the cytoskeleton outside the nucleus. The microtubules in the cytoskeleton facilitate movement of the patches and associated chromosomes, enabling encounters between chromosomes. A protein, dynein, is involved in the separation of mispaired chromosomes. It is also required in the formation of a protein complex between the correctly paired homologous chromosomes.

Which of the following statements are valid conclusions from these findings?

1 The formation of the protein complex between paired homologous chromosomes occurs spontaneously.
2 Mutations in genes coding for protein patches on the nuclear envelope that link the chromosomes to the cytoskeleton inhibit synapsis.
3 Successful formation of the protein complex between paired homologous chromosomes is required for the cell to proceed into metaphase of mitosis.
4 Dynein is necessary to ensure proper synapsis of homologous chromosomes.

A 2, 3 and 4 only

B 1, 2 and 4 only

C 1 and 3 only

D 2 and 4 only

6 The amount of DNA present in a diploid germ cell of 12 chromosomes is 6 picograms (pg). During meiosis I, non-disjunction of a pair of homologous chromosomes occurred.

Which row correctly identifies the amount of DNA and number of chromosomes at different stages of nuclear division?

	Telophase I		Telophase II	
	Amount of DNA (pg) per cell	Number of chromosomes per nucleus	Amount of DNA (pg) per cell	Number of chromosomes per nucleus
A	12	5 or 7	5 or 7	5 or 7
B	12	12	5 or 7	4 or 14
C	6	5 or 7	2.5 or 3.5	5 or 7
D	6	12	2.5 or 3.5	4 or 14

7 Which of the following statement(s) is/are **not** true of the translation process in all eukaryotes?

1 Polypeptides are only synthesised in the cytosol.
2 Amino acids are linked by the formation of peptide bonds catalysed by a ribozyme.
3 Ribosomes contain an amino-acyl tRNA site that is occupied by the initiator tRNA attached to methionine.
4 Amino-acyl tRNA synthetase attaches an amino acid to the 5' end of a tRNA molecule.

A 1, 3 and 4 only

B 2, 3 and 4 only

C 2 and 4 only

D 1 only

8 RNA is involved in the process of protein synthesis. Which of the following descriptions is true about RNA in eukaryotes?

A rRNA, which is coded for by genes found in nucleolus, associates with ribosomal proteins in the cytoplasm to form ribosomal subunits.

B Functional mRNA is formed as a result of post-transcriptional modifications of primary RNA transcript in the nucleus.

C The ribonucleotide sequence of tRNA molecules allows extensive folding and inter-strand complementarity to generate a three-dimensional structure.

D All RNAs must undergo alternative splicing.

9 The template DNA strand for a segment of polypeptide is shown below:

3' ------ GTA ACC GCA TCT CAG AGG ------ 5'

Which of the following will most likely occur if nitrous acid (a mutagenic agent) introduces mutations to this DNA strand by replacing cytosine bases with uracil bases?

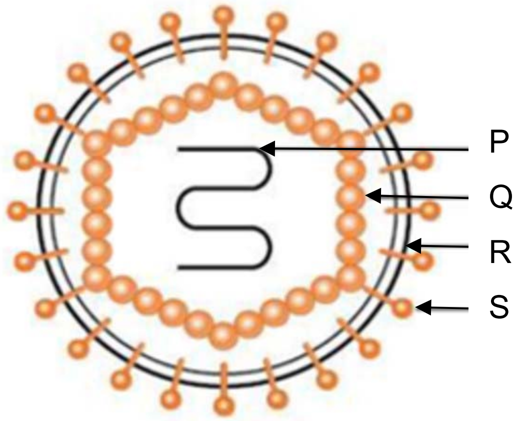
A No polypeptide will be synthesised.

B A truncated polypeptide will be synthesised.

C Four new amino acids with different chemical properties will be found in the polypeptide.

D A polypeptide of original length but with a few new amino acids of different side chains will be synthesised.

10 The diagram shows the structure of a virus.



Which of the following statements are true?

1 P determines the structure of Q and S.
2 Q assists viral entry into the host cell.
3 R and S are required for the entry of the virus into the host cell.
4 Q and R are made of the same components.

A 1 and 2 only

B 1 and 3 only

C 2 and 3 only

D 2 and 4 only

11 How many of these statements about the Human Immunodeficiency Virus (HIV) are correct?

1 The genome is made up of deoxyribonucleotides.
2 The viral enzyme reverse transcriptase is coded for by *pol* gene.
3 Haemagglutinin on viral surface binds to CD4 receptor of helper T cell.
4 HIV enters the host cell via fusion.

A 1

B 2

C 3

D 4

12 The following statements describe the process of conjugation between two bacterial cells.

1 F plasmid replicates semi-conservatively in the donor cell.
2 Replication of F plasmid occurs to form double-stranded DNA in recipient cell.
3 Conjugation tube breaks and retracts.
4 Conjugation tube forms between two bacterial cells.
5 Single-stranded copy of F plasmid is transferred into recipient cell.

Which of the following order describes conjugation correctly?

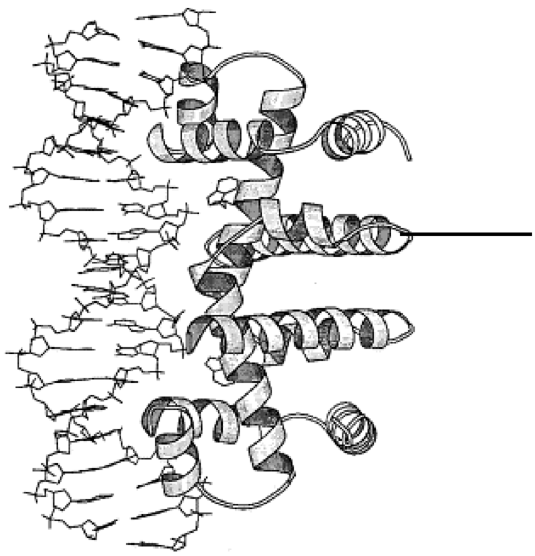
A 1 → 4 → 5 → 2 → 3

B 1 → 5 → 4 → 3 → 2

C 4 → 1 → 5 → 2 → 3

D 4 → 5 → 3 → 1 → 2

13 The diagram shows the binding of an active *trp* repressor protein to DNA.



Which of the options shows the most possible effect of a mutation of the *trp* repressor?

	Part of *trp* repressor affected by mutation	Type of mutation	State of *trp* operon in the presence of tryptophan
A	DNA binding site	Gain-of-function	Transcribed
B	DNA binding site	Loss-of-function	Not transcribed
C	Tryptophan binding site	Loss-of-function	Transcribed
D	Tryptophan binding site	Loss-of-function	Not transcribed

14 The table shows a comparison of some aspects of the genomes and protein-coding genes of eukaryotic and prokaryotic organisms.

Organism	Genome size / base pairs	Chromosome number	Estimated gene number
Human (*Homo sapiens*)	3 billion	46	About 25,000
Mouse (*Mus musculus*)	2.9 billion	40	About 25,000
Fruit fly (*Drosophila melanogaster*)	165 million	8	13,000
Plant (*Arabidopsis thaliana*)	157 million	10	25,000
Roundworm (*Caenorhabditis elegans*)	97 million	12	19,000
Yeast (*Saccharomyces cerevisiae*)	12 million	32	6,000
Bacteria (*Escherichia coli*)	4.6 million	1	3,200

Which of the following statement(s) account(s) for the differences seen in the table?

1 The greater the number of chromosomes an organism has, the larger its genome.
2 The presence of introns in the eukaryotes results in larger genomes and more chromosomes.
3 A larger number of genes would result in a significantly larger genome.
4 *Homo sapiens* and *Mus musculus* are the most closely related, hence they have similar genome size, number of chromosomes and gene number.

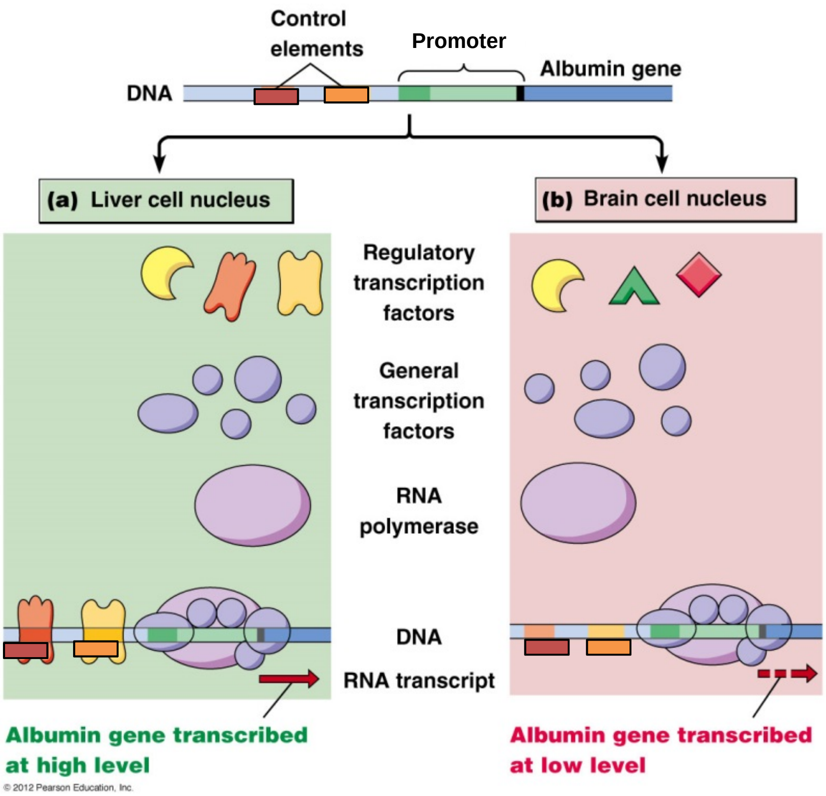
A 2, 3 and 4 only

B 1 only

C 4 only

D None of the above

15 Gene expression of albumin gene is regulated by two control elements and its promoter. These control elements are recognised by regulatory transcription factors which bind to allow for high rate of transcription of the albumin gene.



Which of the following is a result of differential albumin gene expression in liver cells and brain cells?

A Liver and brain cells are differentiated from different pluripotent stem cells, hence they contain different control elements which result in differential gene expression.

B Brain cells contain different RNA polymerases and general transcription factors resulting in low transcription of the albumin gene.

C Brain cells do not contain the regulatory transcription factors that are required to bind to the control elements of the albumin gene to promote the assembly of the transcription complex.

D Liver and brain cells contain the same regulatory control elements, RNA polymerase and transcription factors but a mutation has occurred in the regulatory control elements of the brain cells hence making them dysfunctional.

16 Which of the following is an example of translational control of gene expression?

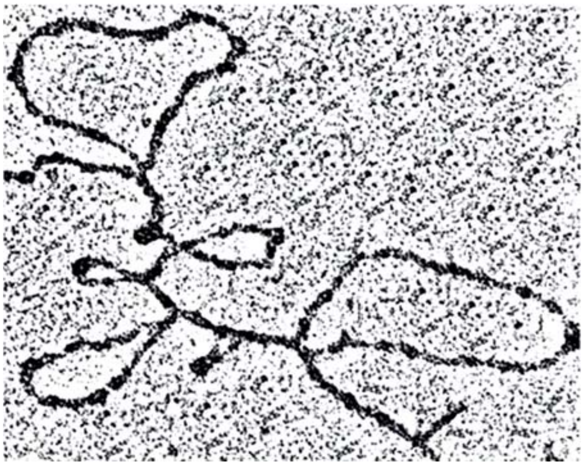
A The binding of protein factors to mRNA to prevent the binding of the small ribosomal subunit

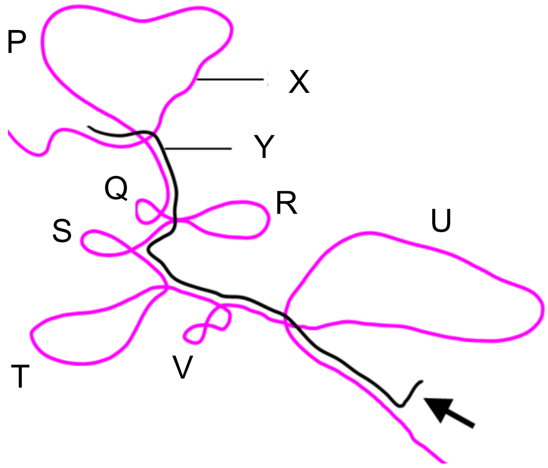
B The activation of proteins by association with other proteins

C The addition of chemical groups such as phosphates to free amino acids

D The degradation of a protein by proteasome

17 The ovalbumin gene from chicken was isolated and made single-stranded and then subsequently mixed together with its mature mRNA. The results were observed under an electron microscope. The electron micrograph and its corresponding diagrammatic representation show the binding of the mRNA to certain regions of the DNA.





Which of the following statement(s) can be concluded?

1 X is the template strand of DNA and Y is the mRNA strand transcribed from X.
2 P, Q, R, S, T, U and V correspond to the introns on the DNA that have been excised from the mRNA.
3 The arrow indicates the 3’ end of the mRNA where the poly(A) tail was added during post-transcriptional modification.
4 The 3’ end of the mRNA is free because there is no corresponding stretch on the template DNA where complementary base pairing can take place.

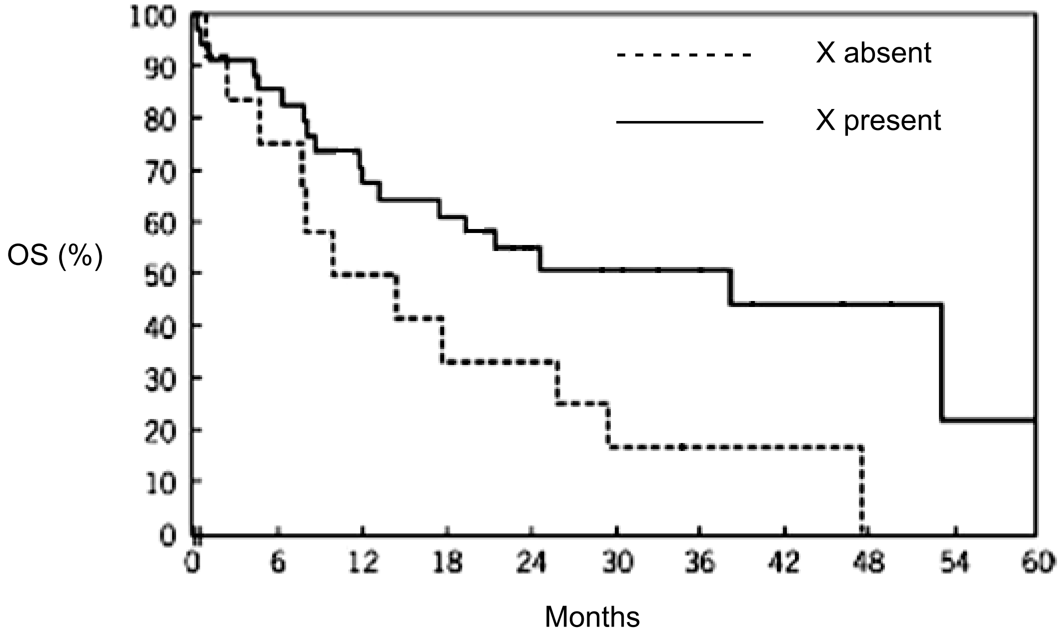
A 1, 2, 3 and 4

B 2, 3 and 4 only

C 1 and 2 only

D 1 only

18 A group of scientists discovered a novel protein and named it X. X is implicated in chromatin structure rearrangement in mammalian cells. The figure shows the overall survival (OS) percentage of cancer patients in the absence and presence of X.



Which of the following statement can best account for the increased survival rates of the cancer patients?

A X is involved in histone acetylation which results in the chromatin having a less compact structure leading to increased expression of tumour suppressor genes, hence allowing cell division to be regulated.

B X is involved in histone acetylation which results in the chromatin having a less compact structure leading leading to expression of oncogenes, hence allowing cell division to be regulated.

C X is involved in histone deacetylation which results in the chromatin having a less compact structure leading to expression of the telomerase gene, hence allowing cell division to be regulated.

D X is involved in histone deacetylation which results in chromatin having a more compact structure leading to a lack of expression of genes involved in angiogenesis, hence allowing cell division to be regulated.

19 In Shorthorn cattle, the allele for the absence of horns is dominant to the allele for the presence of horns. Coat colour can be red (genotype: $\mathbf{C^RC^R}$), roan (genotype: $\mathbf{C^RC^W}$) or white (genotype: $\mathbf{C^WC^W}$).

A roan bull, heterozygous for the hornless trait, is crossed with a cow of the same genotype. Which of the following statement(s) regarding the F_1 offspring is/are true?

1 The probability that a calf from this cross would have the same phenotype as its parents is 3/8.
2 The ratio of horned to hornless calves is 3:1.
3 The number of red and white calves is more than that of roan calves.

A 1 and 2 only

B 2 and 3 only

C 1 only

D 3 only

20 Wing size in *Drosophila* is controlled by a gene with three alleles. The normal wings are long while the other two traits arise as a result of mutation in the same gene locus. The order of dominance for these alleles is as follows.

Long (L) > Vestigial (L^{vg}) = Antlered (L^a)

How many different genotypes for wing size are possible?

A 3

B 4

C 6

D 8

21 The coat colour of Norwegian cattle is mainly determined by the distribution of two pigments: red and black. Both pigments are produced by the action of the enzyme tyrosinase in cells called melanocytes. A low level of activity of the enzyme leads to the production of red pigment, whilst a high activity allows only black pigment production. The activity of the enzyme is increased by the melanocyte stimulating hormone (MSH), which binds to a MSH receptor. The receptor is coded for by the **E** gene locus, which has two alleles, **E** and **e**. No receptor is produced by the recessive allele **e**.

The dominant allele of a second gene, the **A** locus, codes for a protein which binds to and blocks the MSH receptors, thus preventing stimulation of tyrosinase activity in a melanocyte.

Which of the following statements about the two genes and their effects in the colouration of Norwegian cattle is true?

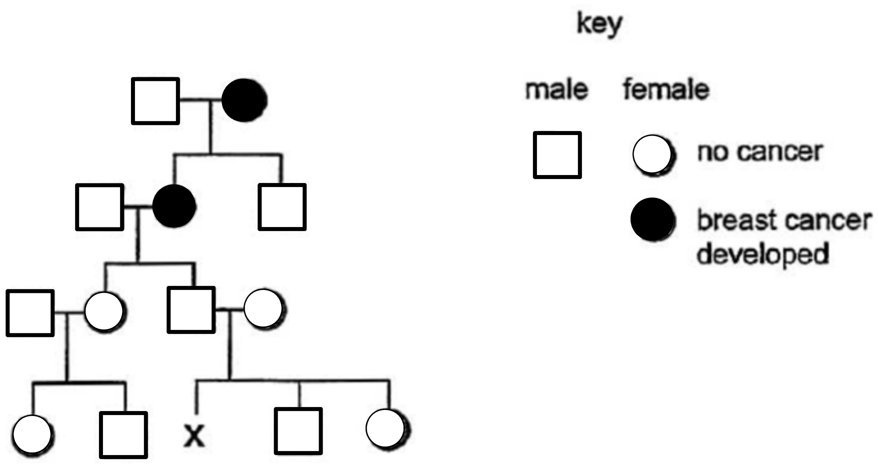
A Allele A is completely epistatic to allele a and allele E is completely epistatic to allele e.

B Cattle with the genotype AAEE have red coats.

C Cattle with black coats must have the genotype aaEe only.

D Cattle with the genotypes aaEE, aaEe and Aaee will have high tyrosinase activity.

22 The diagram below shows the inheritance of a form of breast cancer associated with the presence of just one mutant allele of an autosomal gene *BRCA1*.



What is the probability that woman X inherits the *BRCA1* mutant allele associated with breast cancer?

A 0.00

B 0.25

C 0.50

D 1.00

23 The table below shows the results of an early investigation into the genetic control of phenotypic variation. The dry masses of 5493 bean seeds collected from many plants were classified into nine categories.

Mass of bean/mg	51-150	151-250	251-350	351-450	451-550	551-650	651-750	751-850	851-950
Number of beans	5	38	370	1676	2255	928	187	32	2

Which statement correctly describes these data and could account for the variation shown?

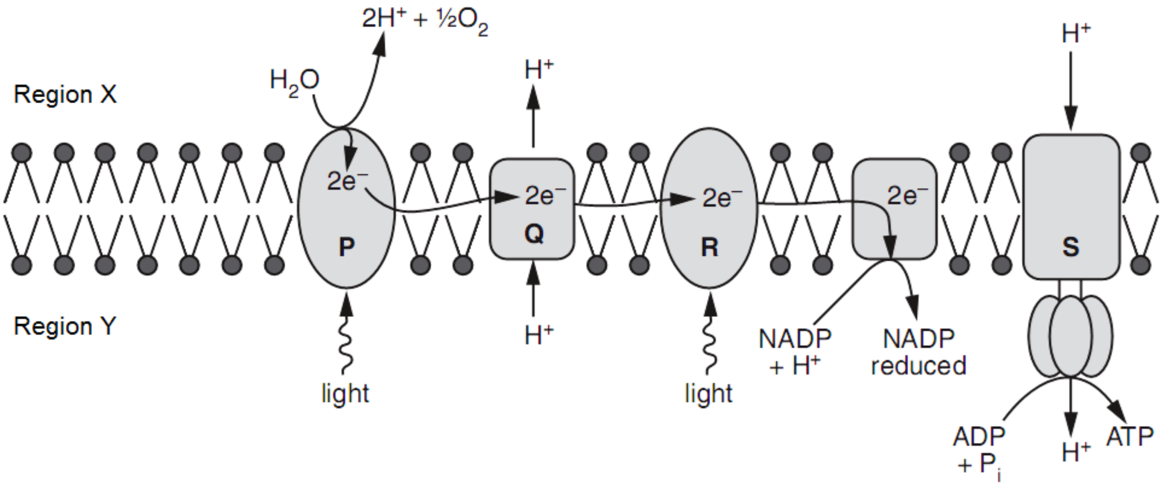
A The phenotypic variation is continuous and could be the result of two non-linked genes acting on their own.

B The phenotypic variation is continuous and could be the result of several non-linked genes acting on their own.

C The phenotypic variation is discontinuous and could be the result of two linked genes acting on their own.

D The phenotypic variation is discontinuous and could be the result of several linked genes acting on their own.

24 The diagram below represents a cross section of a thylakoid, showing some components which are involved in the light-dependent stage of photosynthesis.



Which of the following statements about the following components in the light-dependent stage is true?

A An inhibitor which blocks electron flow through R would inhibit the production of oxygen at P.

B In structures P and R, electrons are passed from one pigment molecule to another until it reaches chlorophyll *a*.

C Region X is expected to have a higher pH than Region Y.

D There is a non-cyclical flow of electrons through structures P, Q, R and S.

25 In respiration, the enzyme hexokinase uses ATP to transfer a phosphate group to glucose to form glucose-6-phosphate.

If a cell only has glucose available for energy and the activity of hexokinase is suddenly inhibited in this cell, which of the following will occur?

1 The cell will not be able to produce pyruvate through glycolysis.
2 Respiratory processes in the mitochondria would not proceed.
3 The use of oxygen by the cell will decrease.

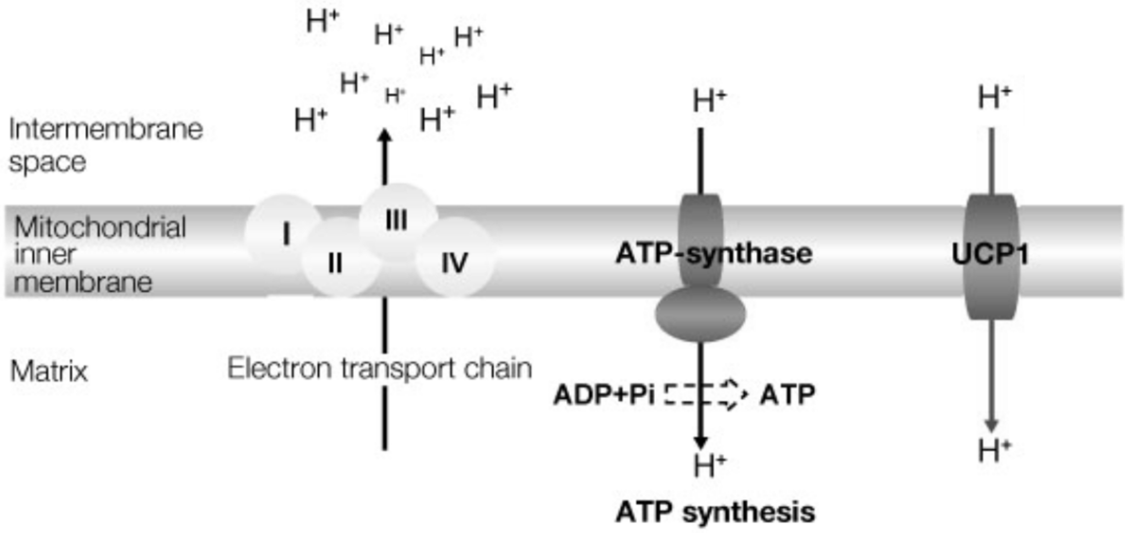
A 1, 2 and 3

B 1 and 2 only

C 1 and 3 only

D 2 and 3 only

26 Thermogenesis is a process that helps certain animals to maintain a constant body temperature. Such animals are found to contain a lot of mitochondria which have proton channels known as UCP1 embedded in the inner membrane as shown below.



Based on the above information, which of the following statements is a likely explanation for the role of UCP1 in thermogenesis?

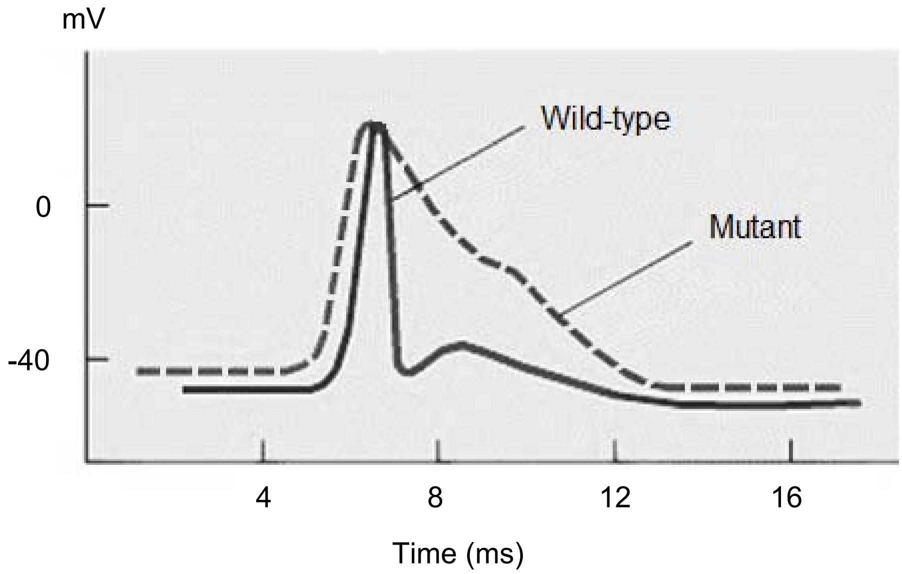
A UCP1 disrupts the flow of electrons along the electron transport chain, channelling protons through it instead of ATP synthase, thus producing heat in the process.

B A proton gradient cannot be established as UCP1 allows protons to pass through the inner membrane passively, hence the energy released from electron transfer is used for heat production.

C The proton motive force is dissipated as heat due to protons flowing through UCP1 instead of passing through ATP synthase.

D The presence of UCP1 allows more protons to diffuse into the intermembrane space so that more protons can eventually diffuse through ATP synthase for ATP production to generate heat.

27 An experiment was carried out to investigate the effect of a particular gene mutation. The action potentials of wild-type flies and mutant flies are shown below.



Which of the following can explain the shape of the action potential in the mutant flies?

A Voltage-gated sodium ion channels were unable to close

B Defective voltage-gated potassium ion channels

C Hyperactive sodium-potassium pumps

D Slow-opening ligand-gated potassium ion channels

28 Some receptors for growth factors activate a protein kinase cascade, usually with the participation of multiple enzymes which cause changes in gene expression. Which of the following statements regarding cell signalling are true?

1 Multiple steps allow the amplification of a signal.
2 External signals can lead to changes in gene expression.
3 The same signal can lead to different responses in cells due to the presence of different target proteins.
4 All cascade systems modify gene expression by activating kinases that enter the cell nucleus by phosphorylating specific transcription factors.

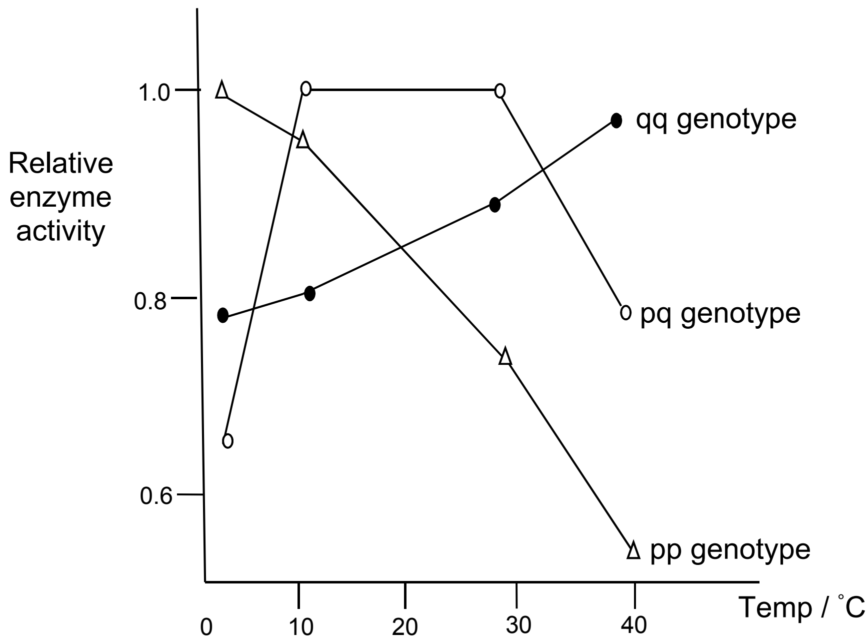
A 1, 2, 3 and 4

B 1, 2 and 3 only

C 1 and 2 only

D 3 and 4 only

29 In the North American catfish *Catostomus elarki*, two alleles represented by p and q, control the synthesis of a vital enzyme. The three possible genotypes (pp, pq, qq) lead to the synthesis of variations of the same enzyme with different temperature optima as shown in the graph below.



When the mean annual temperature is 5°C, which of the following statements are correct?

1 Allele q will be positively selected for.
2 The proportion of allele p in the gene pool will increase over time.
3 The heterozygotes will have a selective advantage over the homozygotes.
4 The catfish will develop a new enzyme variant that has a temperature optimum at 5°C.
5 Catfish with genotype pp will have a selective advantage over the others.

A 1 and 3 only

B 2 and 4 only

C 2 and 5 only

D 3 and 4 only

30 A large population of a certain species of freshwater fish lives in a South America lake. Assuming that there are no mutations, all immigration into the population is prevented and there is no change in selection pressure, which one of the following statements best expresses the probable future of the population?

A All evolution will promptly cease because without mutation, there will be no raw material for evolution.

B The population will begin to decrease in size after three to four generations because of excessive inbreeding that will result from the absence of immigration.

C The population will continue to evolve as selection acts on the different allelic combinations formed during meiosis.

D The population will cease to evolve and may survive for a long time as long as there is no selection.

31 When organochlorine insecticides such as DDT were in widespread use, mosquitoes in malarial regions developed resistance more rapidly than did houseflies in Britain. What could account for the difference in the rates of the development of resistance?

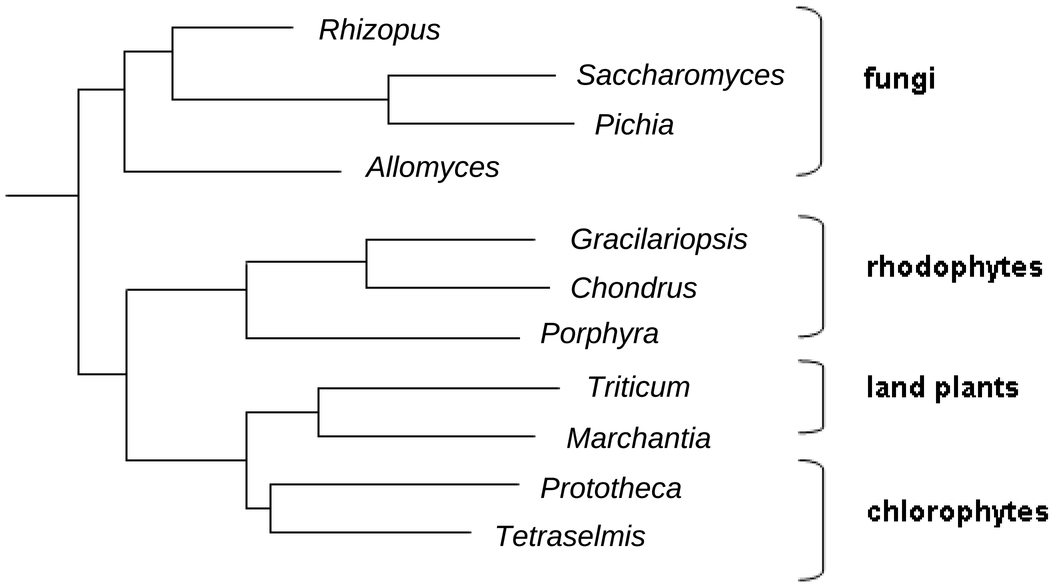
A Mosquitoes produce fewer generations a year.

B More insecticide was used in Britain.

C More insecticide was used in malarial regions.

D Mosquitoes show fewer random mutations per generation.

32 The phylogenetic tree below is derived from comparisons made with mitochondrial DNA from animals, fungi, rhodophytes (red algae) and plants.



What may be concluded from the above phylogenetic tree?

1 *Triticum* and *Marchantia* form a clade while *Allomyces* does not belong to any clade.
2 *Gracilariopsis*, *Chondrus* and *Porphyra* have evolved from the same most recent ancestor.
3 Chlorophytes and land plants are more closely related compared to chlorophytes and rhodophytes.
4 The rhodophytes share a common ancestry with chlorophytes and land plants.

A 1, 2 and 3 only

B 2, 3 and 4 only

C 2 and 4 only

D 3 and 4 only

33 In genetic engineering, which of the following are possible reasons for the limit on the size of the gene to be inserted into a plasmid vector?

1 cDNA is usually used instead of genomic DNA.
2 Probability of number of errors during replication increases as size of gene increases.
3 Number of ligases needed increases with an increase in the size of gene.
4 Efficiency of transformation of competent bacteria decreases with an increase in size of gene.

A 1, 2 and 4 only

B 2, 3 and 4 only

C 1 and 3 only

D 2 and 4 only

34 Which of the following statements regarding Restriction Fragment Length Polymorphisms (RFLP) and their analyses are correct?

1 RFLPs tightly linked to a gene coding for a disease can be used for disease detection.
2 All RFLPs exist as dominant and recessive alleles.
3 RFLPs can identify all single base pair changes in a chromosome.
4 All changes in restriction enzyme sites can be used as genetic markers.

A 1, 2, 3 and 4

B 1, 2 and 3 only

C 2, 3 and 4 only

D 1 and 4 only

35 Which of the following is an ethical concern of the Human Genome Project?

A Difficult to develop treatment for diseases involving multiple genes

B Costly procedures limit genetic testing to those who can afford them

C Genetic testing may not provide reliable and accurate information

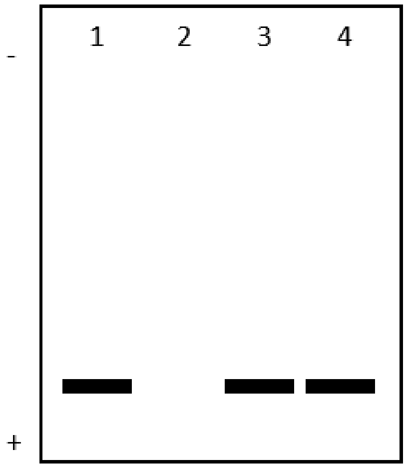
D Unborn foetuses detected with diseases may be aborted

36 Polymerase chain reactions (PCRs) were carried out on fruit fly genomic DNA. The DNA was added to four test-tubes and the treatments for the test-tubes are shown in the table below. The primers were designed to amplify a DNA section which is about 2 kb long.

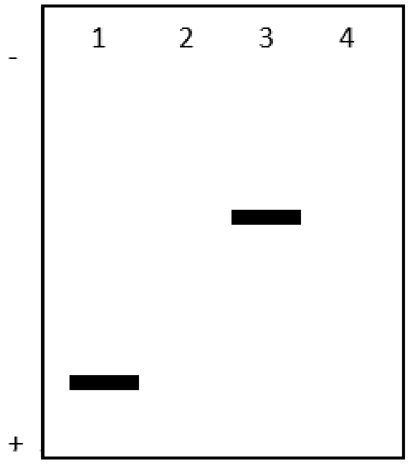
Test-tube	Reagents	Temperature
1	Forward and reverse primers, deoxyribonucleotides, *Taq* polymerase	94°C (for 30 s) → 55°C (for 45 s) → 72°C (for 120 s)
2	Forward and reverse primers, deoxyribonucleotides, *Taq* polymerase	55°C (for 75 s) → 72°C (for 120 s)
3	Forward primers only, deoxyribonucleotides, *Taq* polymerase	94°C (for 30 s) → 55°C (for 45 s) → 72°C (for 120s)
4	Forward and reverse primers, deoxyribonucleotides, *Taq* polymerase	94°C (for 30 s) → 55°C (for 45 s) → 37°C (for 120 s)

After the above treatments were completed, gel electrophoresis was carried out on the contents of each test-tube. Which of the following electrophoregrams shows the correct results for the tubes?

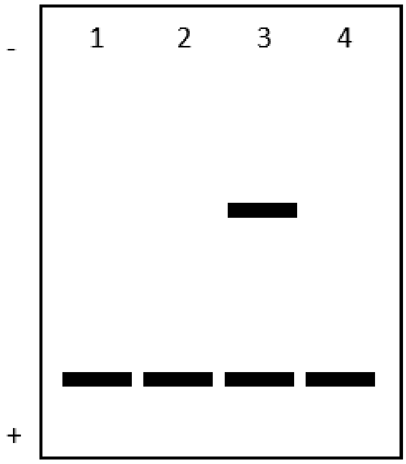
A



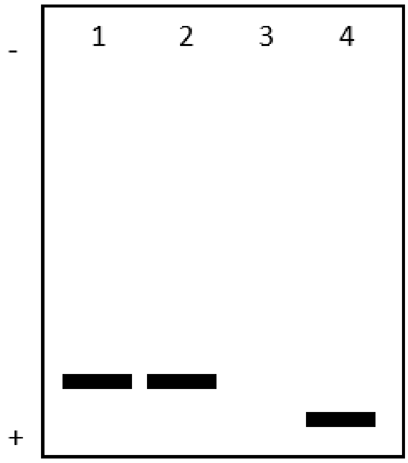
B



C



D



37 Which of the following best illustrates totipotency?

A A somatic cell isolated from a root tip develops into a normal adult plant.

B Stem cells are able to divide indefinitely.

C Mesenchymal stem cells can differentiate into an extensive range of cell types, including bone cells, cartilage cells, muscle cells and fat cells.

D The replacement of the nucleus of an unfertilised egg with that of a pancreatic cell converts the egg into a pancreatic cell.

38 Which of the following are social implications for the use of gene therapy in treating genetic diseases?

1 Gene therapy might provide alternative treatments for patients where conventional treatments have failed.
2 Genetic enhancements can be costly and accessible only to the wealthy.
3 There is difficulty in determining which conditions are normal and which are considered disorders.

A 1, 2 and 3

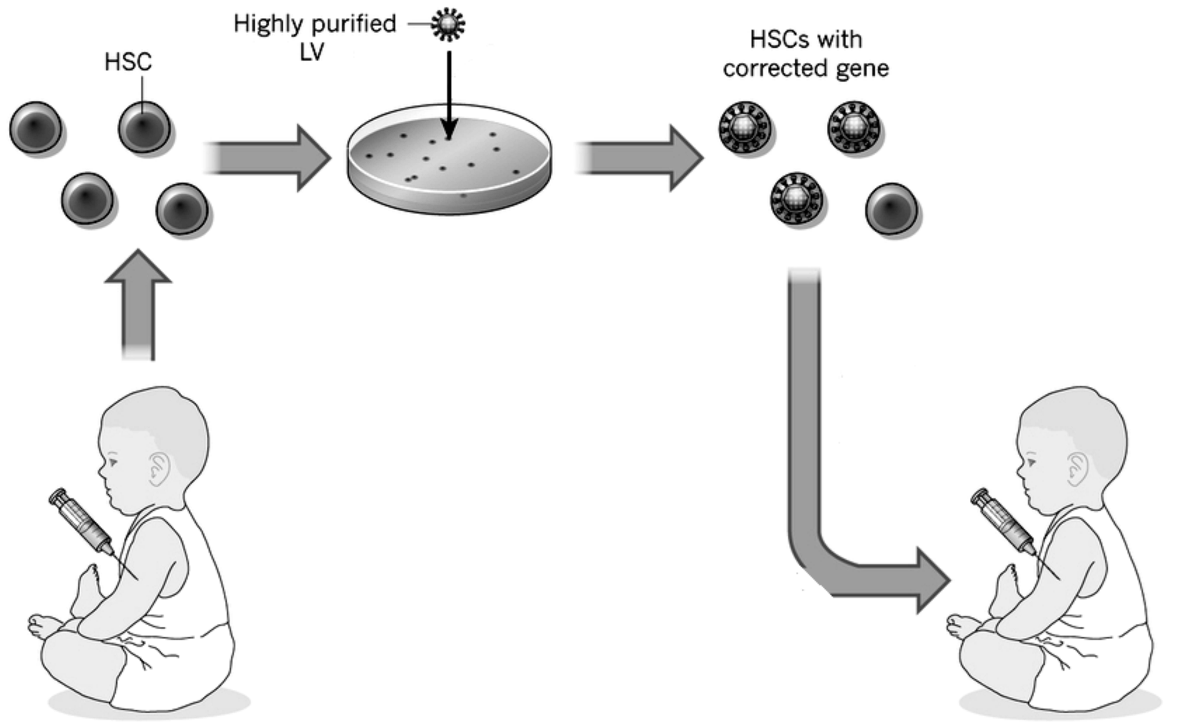
B 1 and 2 only

C 1 and 3 only

D 2 and 3 only

39 Metachromatic leukodystrophy (MLD) is an inherited disorder caused by a deficiency in arylsulphatase A (ARSA) enzyme activity in leukocytes. Patients with MLD accumulate a toxic metabolite and die within a few years.

In a clinical trial, a team of scientists collected haematopoietic stem cells (HSCs) from the bone marrow of children with MLD and exposed them to lentiviral vectors (LV) carrying normal ARSA genes. These genes were then integrated into HSC genomic DNA. HSCs with the corrected gene were then re-introduced into the children's bone marrow.



Which of the following statement(s) regarding the treatment of MLD is/are true?

1 This method of treatment is beneficial as it reduces the risk of incompatibility of HSC transplants.
2 This method of treatment is less effective than introducing lentiviruses containing the normal ARSA genes into the patient directly.
3 Other than HSCs, it is also possible to use leukocytes as target cells for gene therapy.

A 1, 2 and 3

B 1 and 2 only

C 1 and 3 only

D 3 only

40 Transgenic crops expressing insecticidal toxins could provide an effective means of pest control. However, the widespread cultivation of such transgenic crops is expected to promote the development of toxin-resistant pests, hence eventually compromising the usefulness of the pest management strategy. Two planting strategies have thus been recommended to prevent the development of toxin-resistant pests:

Strategy 1: Separate fields of transgenic plants and non-transgenic plants are planted

Strategy 2: ‘Seed mixtures’ of such transgenic plants and non-transgenic plants in the same field are planted

Which of the following considerations would most likely encourage farmers to favour Strategy 1 over Strategy 2?

A Low mortality of susceptible insects on toxin-free plants

B Movement of randomly mating insects from plant to plant within a field

C Concern that 'superweeds' might emerge in fields with 'seed mixtures'

D When toxin resistance is recessive and frequency of recessive alleles is low